

Application and extensions Meta-analysis of single-case experimental studies

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Applications and extensions

1. Network meta-analysis (cfr. other presentation)
2. Meta-analysis of single-case experimental studies

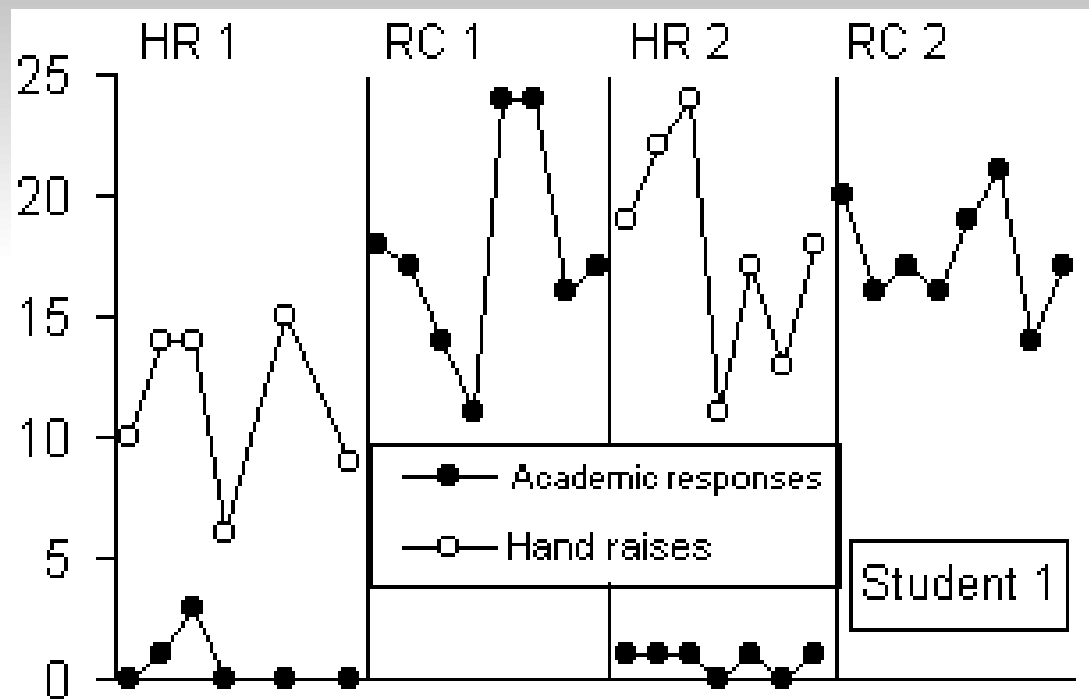
Van den Noortgate, W., & Onghena, P. (2007). The aggregation of single-case results using hierarchical linear models. *The Behavior Analyst Today*, 8, 196-209.



Example 1: stimulating response behavior in the classroom

Narayan, J. S., Heward, W. L., Gardner, R., Courson, F. H., Omness, C. K.
(1990), *JABA*, 23, 483-490.





Questions:

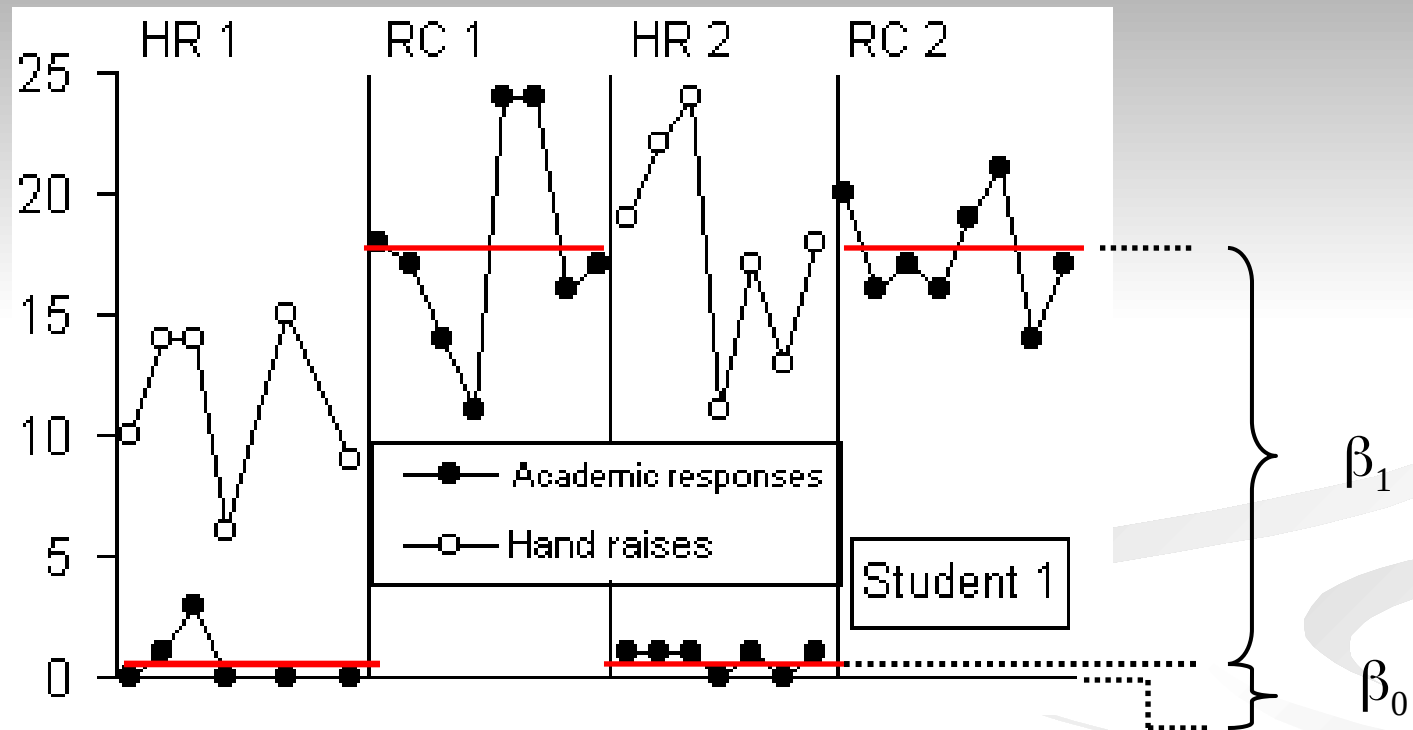
1. Is the effect statistically significant?
2. How large is the effect?



Measuring and testing the effect

$$Y_i = \beta_0 + \beta_1 (\text{Treatment})_i + e_i \quad \text{met } e_i \sim N(0, \sigma_e^2)$$





Effect size

$$\beta_1 = \mu_B - \mu_A$$
$$\delta_{SC} = \frac{\beta_1}{\sigma_e} = \frac{\mu_B - \mu_A}{\sigma_e}$$

$$g_{SC} = \frac{b_1}{s_p} = \frac{\bar{X}_B - \bar{X}_A}{s_p}$$



Questions:

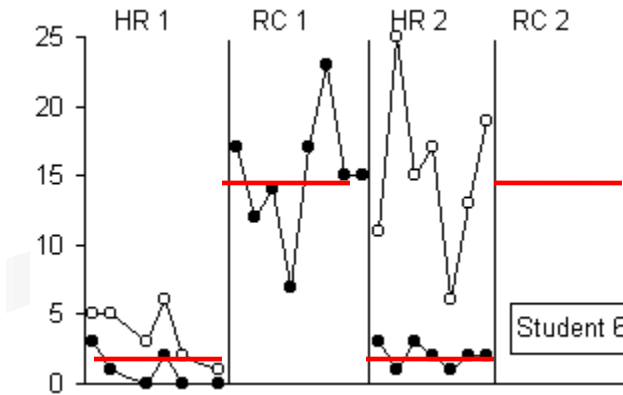
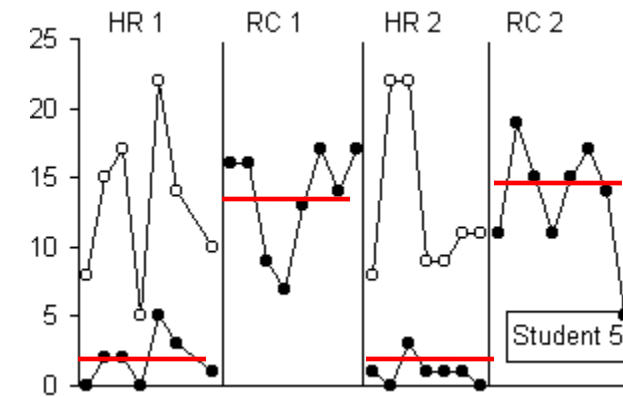
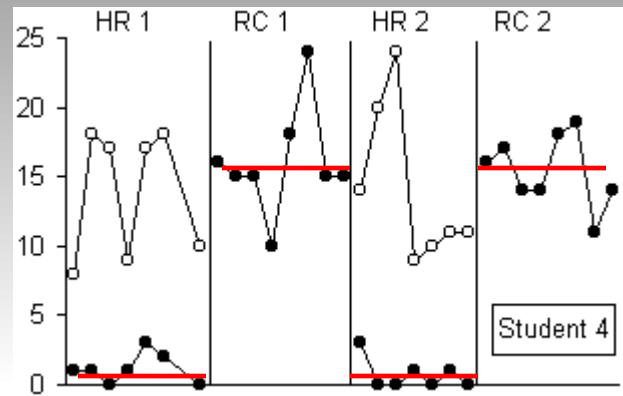
1. Is the effect statistically significant?
2. How large is the effect?

Problem of generalizability

⇒ replication (direct / systematic; within / over studies)

3. Combining results over cases?





Level 1 (within students):

$$Y_{ij} = \beta_{0j} + \beta_{1j} (\text{treatment})_{ij} + e_{ij} \quad \text{met } e_{ij} \sim N(0, \sigma_e^2)$$

Level 2 (between students):

$$\begin{aligned} \beta_{0j} &= \gamma_{00} + u_{0j} \\ \beta_{1j} &= \gamma_{10} + u_{1j} \end{aligned}$$

$$\text{met } \mathbf{u} \sim N(\mathbf{0}, \mathbf{\Omega})$$

Results:

Fixed coefficients:

Intercept	γ_{00}	1.04	(0.31)
Treatment	γ_{10}	14.33	(0.74)

Variance components:

Level 2:

Intercept	σ^2_{u0}	0	
Treatment	σ^2_{u1}	2.17	(1.68)

Level 1:

Intercept	σ^2_i	7.40	(0.84)
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Example 2: The effect of choice-making as an intervention for problem behavior: A meta-analysis

Shogren, K.A., Fagelly-Luby, M.N., Bae, S.J., & Wehmeyer, M.L. (2004), *Journal of Positive Behavior Interventions*, 6, 228-237.



A three level model

Level 1:
$$Y_{ijk} = \beta_{0jk} + \beta_{1jk} \textit{Choice}_{ijk} + r_{ijk}$$

Level 2:
$$\beta_{0jk} = \theta_{00k} + u_{0jk}$$

$$\beta_{1jk} = \theta_{10k} + u_{1jk}$$

Level 3:
$$\theta_{00k} = \gamma_{000} + v_{00k}$$

$$\theta_{10k} = \gamma_{100} + v_{10k}$$



Note:

- To make data comparable:
standardization



Fixed

Intercept	3.07	(0.36)
Choice	-1.84	(0.46)

Variances

Level 3:

Intercept	0	
Choice	0.33	(0.81)
Covariance	0.013	(0.40)

Level 2:

Intercept	2.45	(0.84)
Choice	3.07	(1.21)
Covariance	-2.02	(0.88)

Level 1:

Intercept	1.00	(0.063)
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A three level model

Level 1:
$$Y_{ijk} = \beta_{0jk} + \beta_{1jk} \textit{Choice}_{ijk} + r_{ijk}$$

Level 2:
$$\beta_{0jk} = \theta_{00k} + \theta_{01} Z_{jk} + u_{0jk}$$

$$\beta_{1jk} = \theta_{10k} + \theta_{11} Z_{jk} + u_{1jk}$$

Level 3:
$$\theta_{00k} = \gamma_{000} + \gamma_{001} W_k + v_{00k}$$

$$\theta_{10k} = \gamma_{100} + \gamma_{101} W_k + v_{10k}$$



Fixed

Intercept	3.07	(0.36)	3.34	(1.07)
Choice	-1.84	(0.46)	-2.86	(1.40)
Male			-0.82	(0.76)
Order			0.37	(1.02)
Male x Choice			0.33	(0.91)
Order x Choice			0.98	(1.30)

Variances

Level 3:

Intercept	0	0
Choice	0.33 (0.81)	0.48 (0.92)
Covariance	0.013 (0.40)	-0.15 (0.39)

Level 2:

Intercept	2.45 (0.84)	2.59 (0.93)
Choice	3.07 (1.21)	3.28 (1.35)
Covariance	-2.02 (0.88)	-2.17 (0.99)

Level 1:

Intercept	1.00 (0.063)	1.00 (0.063)
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Alternatively: meta-analysis of effect sizes

Level 1:
$$Y_{ijk} = \beta_{0jk} + \beta_{1jk} \textit{Choice}_{ijk} + r_{ijk}$$

Level 2:
$$\beta_{0jk} = \theta_{00k} + u_{0jk}$$

$$\beta_{1jk} = \theta_{10k} + u_{1jk}$$

Level 3:
$$\theta_{00k} = \gamma_{000} + v_{00k}$$

$$\theta_{10k} = \gamma_{100} + v_{10k}$$



Alternatively: meta-analysis of effect sizes

Level 1: $g_{jk} = \beta_{1jk} + e_{jk}$

Level 2: $\beta_{0jk} = \theta_{00k} + u_{0jk}$

$$\beta_{1jk} = \theta_{10k} + u_{1jk}$$

Level 3: $\theta_{00k} = \gamma_{000} + v_{00k}$

$$\theta_{10k} = \gamma_{100} + v_{10k}$$

Alternatively: meta-analysis of effect sizes

Level 1: $g_{jk} = \beta_{1jk} + e_{jk}$

Level 2:

$$\beta_{1jk} = \theta_{10k} + u_{1jk}$$

Level 3:

$$\theta_{10k} = \gamma_{100} + v_{10k}$$

Fixed

Intercept

3.07 (0.36)

Choice

-1.84 (0.46)

-1.81 (0.49)

Variances

Level 3:

Intercept

0

Choice

0.33 (0.81)

0.60 (1.20)

Covariance

0.013 (0.40)

Level 2:

Intercept

2.45 (0.84)

Choice

3.07 (1.21)

2.93 (1.28)

Covariance

-2.02 (0.88)

Level 1:

Intercept

1.00 (0.063)



Extensions ...

- Autocorrelation
- Other covariates, e.g. for 2 treatments
- Linear trends:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 T_i + \beta_3 D_i T_i + e_i$$

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 T_i + \beta_3 D_i T_i + e_i$$

$$= (\beta_0 + \beta_1 D_i) + (\beta_2 + \beta_3 D_i) T_i + e_i$$

